FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST –X
PAPER –1
TEST DATE: 21-05-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

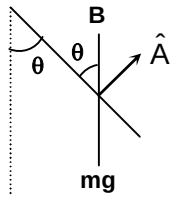
Section – A

1. B, D

Sol. If $E_2 > E_1$; $E_1 < \dfrac{E_1r_2 + E_2r_1}{r_1 + r_2} < E_2$

2. B, D

Sol. Applying Energy conservation, initially, kinetic energy = 0



gravitational P.E. = 0 (say) &

Magnetic P.E. = 0

where, μ = magnetic moment of the loop

$= i \cdot \left(\dfrac{\sqrt{3}a^2}{4}\right)$

Finally, when the loop becomes horizontal, Kinetic energy = 0 gravitational P.E. = $mg\left(\dfrac{a}{\sqrt{3}}\right)$, (because mg acts on the centre of mass)

magnetic P.E. = –μB

$\Rightarrow \quad 0 + 0 + 0 = 0 + \dfrac{mga}{\sqrt{3}} - \mu B$

$\Rightarrow \quad B = \dfrac{4mg}{3ia}$

for equilibrium position, torque due to magnetic force should be balanced by the torque due to gravitational force (mg).

$\mu B \sin(90 - \theta) = \dfrac{mg \sin\theta a}{\sqrt{3}}$

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3. A, C, D

Sol. (A) E at C due to outside charges = 0

E at C due to induced $= \frac{K2q}{(2R)^2} = \frac{Kq}{2R^2}$

(C) $\frac{2Kq}{2R} + \frac{Kq'}{R} = 0$

$q' = -q$

(D) $\Delta Q = 4q - (-q) = 5q$

4. A, B, C

Sol. Just after the switch is closed, flux through the inductor will remain unchanged.

5. B, D

Sol. $V_L^2 + V_R^2 = V_R^2 + V_C^2$

$V_L = V_C$

Circuit in resonance

$\therefore V_2 = 400$ volt

$400 = 10 \times 2R$

$\therefore R = 20\Omega$

6. A, B, D

Sol. $\frac{1}{2}k\ell^2 = 25$ J

Work done by the gas, $\Delta W = P_{atm}A\ell + \frac{1}{2}k\ell^2 = \frac{3}{2}k\ell^2 = 75J$

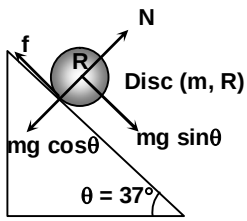
$\Delta U = nC_V\Delta T = n\frac{3}{2}R\Delta T = \frac{3}{2}nR(4T_0 - T_0) = \frac{9}{2}nRT_0$

$= \frac{9}{2}P_{atm}A\ell = \frac{9}{2}k\ell^2 = 9 \times 25 = 225J$

$\Delta Q = \Delta W + \Delta U = 75 + 225 = 300$ J

7. C

Sol. $f = \frac{mg\sin\theta}{3}$ (pure rolling)



$f \le \mu N$

$\frac{mg\sin\theta}{3} \le \mu mg\cos\theta$

$\frac{\tan\theta}{3} \le \mu$ (pure rolling)

$\mu \ge \frac{1}{4}$ (pure rolling) and $a = \frac{2g\sin\theta}{3} = \frac{2}{3} \times 10 \times \frac{3}{5} = 4\ m/s^2$

For $\mu < \frac{1}{4} \Rightarrow$ kinetic friction will act

$a_{cm} = g(\sin\theta - \mu\cos\theta)$

$f = \mu mg\cos\theta$

8. A

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9. B

10. A

Sol. $\varepsilon_{OA} = \frac{1}{2}B\omega(2L^2 - 0) = B\omega L^2$

$\varepsilon_{OD} = \frac{1}{2}B\omega(2L^2 - 0) = B\omega L^2$

$\varepsilon_{CD} = \frac{1}{2}B\omega(2L^2 - L^2) = \frac{1}{2}B\omega L^2$

$\varepsilon_{AD} = \frac{1}{2}B\omega(2L^2 - 2L^2) = 0$

Section – B

11. 0.04

Sol. $\frac{dQ}{dt} = nC_P\frac{dT}{dt}$

$q = \frac{5}{2}nR\frac{dT}{dt}$

$nR\frac{dT}{dt} = \frac{2q}{5}$ ...(i)

$P\frac{dV}{dt} = nR\frac{dT}{dt}$

$PA\frac{dx}{dt} = \frac{2q}{5}$

$v = \frac{2q}{5PA}$

$v = \frac{2q}{5(P_0A + mg)}$

12. 0.02

Sol. Range = $v_0 \times$ time of flight (t)

$\Rightarrow t = \frac{2v_0^2}{gv_0} = \frac{2v_0}{g}$

If y is the height of balloon at any instant t and $\frac{dy}{dt}$ its velocity then

$y = -\left(\frac{dy}{dt}\right)\left(\frac{2v_0}{g}\right) + \frac{1}{2}g\left(\frac{2v_0}{g}\right)^2$

$\Rightarrow \frac{dy}{dt} + \frac{g}{2v_0}y - v_0 = 0$

$\frac{dy}{\left(v_0 - \frac{g}{2v_0}y\right)} = dt$

$\Rightarrow -\frac{2v_0}{g}\ell n\left(v_0 - \frac{g}{2v_0}y\right) = t + c$

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At t = 0, y = 0 $\Rightarrow$ $c = -\frac{2v_0}{g}\ell n v_0$

Simplifying, we get

$y = \frac{2v_0^2}{g}\left[1 - e^{-gt/2v_0}\right]$

13. 0.25

Sol. The movable pulley moves vertically upwards

Constraint equation

$2\sqrt{x^2 + b^2} + y = \text{length of string} = \text{constant}$

$\therefore$ Differentiating w.r.t. time

$\frac{2}{2\sqrt{x^2+b^2}} \cdot 2x\frac{dx}{dt} + \frac{dy}{dt} = \frac{d\ell}{dt} = 0$

But $\frac{dy}{dt} = v$

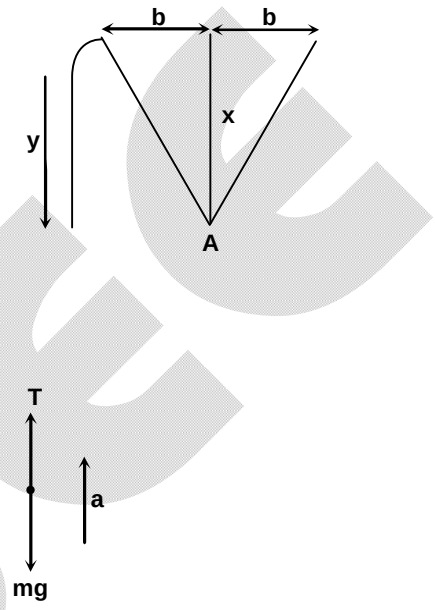
$\therefore \frac{dx}{dt} = -\frac{v}{2x}\sqrt{x^2 + b^2}$

Again differentiating w.r.t. time gives

$\frac{d^2x}{dt^2} = -\frac{b^2v^2}{4x^3}$

Now, considering the F.B.D of m

$T = mg + ma = mg\left(1 + \frac{b^2v^2}{4gx^3}\right)$



14. 2.50

Sol. Let acceleration of cylinder down the incline is a_1 and angular acceleration is α as shown in figure.

$mg\sin\theta + f - T_1 = Ma_1$ ...(i)

$(T_1 + f)R = \frac{MR^2}{2}\alpha$ ...(ii)

Let acceleration of plank down the incline is a

$\therefore$ $Mg\sin\theta - T - f = Ma$ ...(iii)

$T - mg = ma$ ...(iv)

For pure rolling of cylinder over string

$a_1 = R\alpha$ ...(v)

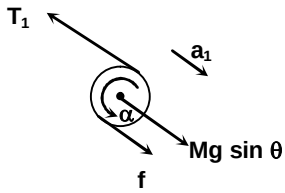
For pure rolling of cylinder above plank

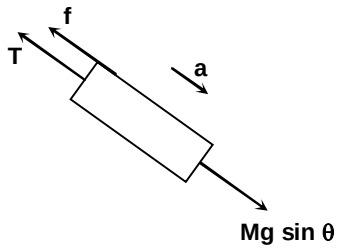
$a = a_1 + R\alpha$ ...(vi)

$\Rightarrow a = 2R\alpha = 2a_1$

Solving above equation we get

$a = \frac{5g}{26}$ and $a_1 = \frac{5g}{52}$





15. 48.00

Sol. Minimum area of a rectangular plane mirror required,

$A = 12 \times 4 = 48\ cm^2$

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16. 0.25

Sol. Let r be the radius of the raindrop when it has fallen a distance x in time t, then mass of the rain drop, $m = \frac{4}{3}\pi r^3\rho$

Where ρ is the density of the water

Since, the rate of increase of volume = μ(surface area)

$\therefore \frac{d}{dt}\left(\frac{4}{3}\pi r^3\right) = \mu(4\pi r^2)$

Or $\frac{dr}{dt} = \mu$

On integrating, $r = \mu t + r_0$

According to the second law of motion

$$\frac{d}{dt}(mv) = mg$$

Or $\frac{d}{dt}\left[\frac{4}{3}\pi(r_0+\mu t)^3\rho v\right] = \frac{4}{3}\pi(r_0+\mu t)^3\rho g$

Or $d\left[(r_0+\mu t)^3 v\right] = (r_0+\mu t)^3 g dt$

Integrating both the sides,

$(r_0+\mu t)^3 v = \frac{(r_0+\mu t)^4 g}{4\mu} + C$

At t = 0, v = 0

$\therefore C = -\frac{r_0^4}{4\mu}g$

Thus, $(r_0+\mu t)^3 v = \frac{(r_0+\mu t)^4 g}{4\mu} - \frac{r_0^4 g}{4\mu}$

Or $v = \frac{g}{4\mu}\left[r_0+\mu t - \frac{r_0^4}{(r_0+\mu t)^3}\right]$

17. 3.00

Sol. When normal reaction will be zero, then acceleration of sphere along downward is g. If acceleration of wedge is A along the plane.

Then $A\sin\alpha = g$

$\Rightarrow A = \frac{g}{\sin\alpha}$

$\therefore F\sin\alpha + Mg\sin\alpha = MA$

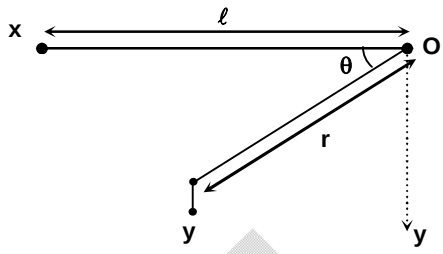
$F\sin\alpha = M(A - g\sin\alpha)$

$F = Mg\cot^2\alpha = 3Mg$

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18. 0.09

Sol. Let the second nail be located at a distance r from the first nail and angle θ below horizontal as shown. Now speed of the bob at its lowest position can be found using conservation of energy.



$$\frac{1}{2}mv^2 = mg[r\sin\theta + \ell - r]$$

$$\Rightarrow v^2 = 2g(r\sin\theta + \ell - r) \quad \ldots(i)$$

The particle will compete circle about the second nail if $v^2 = 5g(\ell - r)$ …(ii)

From (i) and (ii)

$2(r\sin\theta + \ell - r) = 5(\ell - r)$

$\Rightarrow 3r + 2r\sin\theta = 3\ell$ …(iii)

Now coordinates of the nail are

$x = r\cos\theta$, $y = r\sin\theta$

Using these in (iii), we get

$$3\sqrt{x^2 + y^2} + 2y = 3\ell$$

Or $9(x^2 + y^2) = (3\ell - 2y)^2$

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Chemistry

PART – II

Section – A

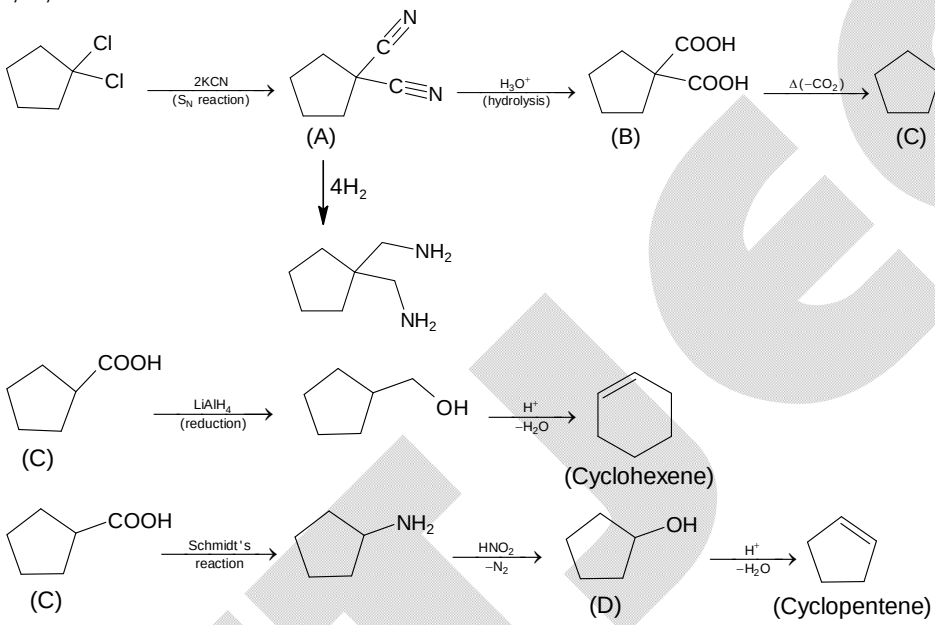
19. A, C

Sol. (A) Because at the required temperature of Lassaigne test, N_2 expels out in the form of a gas in $C_6H_5N_2Cl$.

(C) NH_2OH does not contain carbon.

20. A, B, D

Sol.



21. A, B, C

Sol. (D) is not condensed polymer

22. A, B, C

Sol. For adiabatic process

$\Delta U = W = -P(V_2 - V_1) = -1(20 - 10) = -10$ L atm $= -1013$ J

$\Delta H = C_P \Delta T$

$\Delta U = C_V \Delta T$

$$\frac{\Delta H}{\Delta U} = \frac{C_P}{C_V}$$

As gas is monoatomic,

$$C_P = \frac{5}{2}R \text{ and } C_V = \frac{3}{2}R$$

$$\therefore \frac{\Delta H}{\Delta U} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3}$$

$$\therefore \Delta H = \frac{5}{3}\Delta U = \frac{5}{3}(-1013) = -1688 \text{ J}$$

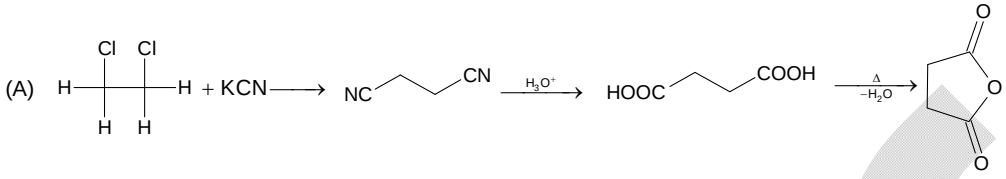
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$$\Delta U = C_V\left(T_2 - T_1\right) = \frac{3}{2}R\left(T_2 - T_1\right) = -1013$$

$\therefore\ T_2 = 218.8\ K$

23. A, B

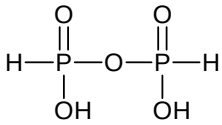
Sol.



(B) $H_3C-CHCl_2 \xrightarrow{KCN} H_3C-CH(CN)_2 \xrightarrow{H_3O^+} H_3C-CH(COOH)_2 \xrightarrow[\text{decarboxylation}]{\Delta} CH_3CH_2COOH$

24. B, C, D

Sol. Pyrophosphorus acid ($H_4P_2O_5$) is dibasic acid as it contains two P – OH bonds, strongly reducing in nature due to the presence of two P – H groups and contains one P – O – P bond.



Hence, (B), (C) and (D) are the correct answers.

25. C

Sol. (i) Non-superimposable mirror image; enantiomer.
(ii) Superimposable mirror image, identical.
(iii) Configuration at one chiral carbon is different; Diasteromers
(iv) Position of – OH is different; positional isomers.

26. D

Sol. (I) $pH = 7 + \frac{1}{2}\left(4.74 + \log\frac{25}{75}\right) = 9.1$

(II) $pH = \frac{6.4 + 10.4}{2} = 8.4$

(III) $pOH = 4.74 + \log\frac{5}{5} = 4.74$

$pH = 14 - 4.74 = 9.26$

(IV) $\alpha = \sqrt{\frac{2\times10^{-5}}{0.1}} = \sqrt{2}\times10^{-2}$

$\left[OH^-\right] = C\alpha = 0.1\times\sqrt{2}\times10^{-2} = \sqrt{2}\times10^{-3}$

$pOH = 3 - \log\sqrt{2} = 2.85$

$\Rightarrow pH = 11.15$

27. C

Sol. (I) Ti^{+3} is a oxidizing agent.

(II) $\left(SiH_3\right)_3N$ has $p\pi - d\pi$ overlap between N and Si.

(III) Sn^{+2} is a reducing agent.

(IV) $\left(CH_3\right)_3N$ is Lewis base.

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28. A

Sol. (I) Markonikov's addition of H_2O : Oxymercuration-demercuration.

(II) Anti- Markonikov's addition of H_2O : Hydroboration.

(III) Epoxidation at more electron rich alkene : mCPBA.

(IV) Epoxidation at less electron rich alkene : H_2O_2 / OH^-

Section – B

29. 164.30

Sol. $HCl(g) + aq. \longrightarrow H^+(aq) + Cl^-(aq)$

$-76.5 = 0 + \Delta H_f\left(Cl^-(aq)\right) + 87.8$

$\Delta H_f\left(Cl^-(aq)\right) = -164.30$

30. 150.00

Sol. Moles of Al^{+3} reduced $= \dfrac{500(0.3-0.2)}{1000} = 0.05$

Number of Faraday = 0.05 ×3

$0.05 \times 3 \times 96500 = 96.5 \times t$

$t = 150$ sec ond

31. 50.00

Sol. $NaHCO_3 \longrightarrow \underset{1}{Na^+} + \underset{1}{HCO_3^-}$

$\underset{1-h}{HCO_3^-} + H_2O \longrightarrow \underset{h}{H_3O^+} + \underset{h}{CO_3^{-2}}$

$i = 1 + h + h + 1 - h = 2 + h$

$0.372 = (2+h) \times 0.08 \times 1.86$

$2 + h = 2.5$

h = 50%

32. 20.00

Sol. Meq. of NaOH $+ \dfrac{1}{2}$ Meq. of $Na_2CO_3 = 15 \times \dfrac{1}{20}$

Meq. of NaOH + Meq. of $Na_2CO_3 = 20 \times \dfrac{1}{20}$

Meq. of NaOH $= (30-20)\dfrac{1}{20}$

$\dfrac{W \times 1000}{40} = 0.5$

$W = 20 \times 10^{-3} = 20$ mg

33. 56.25

Sol. $n = \dfrac{1 \times 1.25}{0.0821 \times 300} = 0.05$

$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$

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$T_2 = 300\left(\frac{1.25}{10}\right)^{\frac{5}{3}-1}$

$= 300\left(\frac{1}{2}\right)^{3\times\frac{2}{3}} = 75$

$\Delta H = nC_P\Delta T$

$= 0.05\times 5\times(300-75)$

$= 56.25$ Cal

34. 9.26

Sol. Volume of unit cell $\left(3\times10^{-8}\right)^3 = 27\times10^{-24}$ cc.

Volume of element weighing 100 gm $= \frac{100}{8} = 12.5$ cc

Number of unit cell present in 100 g of element $= \frac{12.5}{27\times10^{-24}} = 4.63\times10^{23}$

Each BCC unit cell contains two atoms

Number of atoms $= 2\times4.63\times10^{23}$

$= 9.26\times10^{23}$

= 9.26

35. 35.00

Sol. $5CO + I_2O_5 \longrightarrow I_2 + 5CO_2$

$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

25 mm $100\times\frac{1}{2} = 50$ mm

Millimoles of CO = 125

Wt. of CO $= 125\times10^{-3}\times28 = 3.5$

Hence, % of CO $= \frac{3.5\times100\%}{10} = 35\%$

36. 76.03

Sol. $CH_3CHO + CuO \longrightarrow CH_3COOH + Cu_2O$

(Red ppt.)

143.6 gram of Cu_2O is given by 44 gram of acetaldehyde

1.2 gram of Cu_2O is given by $= \frac{44\times1.2}{143.6}$ = 0.368 gram acetaldehyde

% $CH_3CHO = \frac{0.368}{0.484}\times100 = 76.03\%$

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Mathematics

PART – III

Section – A

37. A, B, C

Sol. $(x-6)^4 + (x-8)^4 = 16$. Let $x - 6 = y + 1 \Rightarrow x - 8 = y - 1$

Now, $(y+1)^4 + (y-1)^4 = 16 \Rightarrow y^4 + 6y^2 - 7 = 0 \Rightarrow y^2 + 7 = 0$ and $y^2 - 1 = 0$

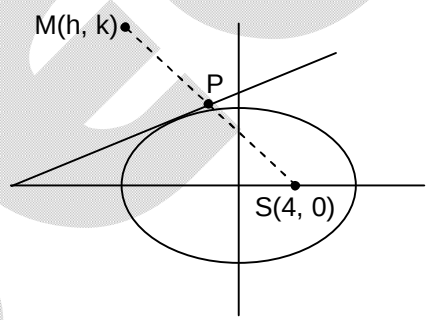
$y = \pm\sqrt{7}\,i$ or $y = \pm 1 \Rightarrow x = 7 \pm 1$ or $x = 7 \pm \sqrt{7}\,i \Rightarrow a = 7, b = 1, c = 7, d = 7$

38. A, C, D

Sol. $\left(x + y = \frac{\pi}{2}, \; xy = 1\right)$ and $\left(x + y = \frac{5\pi}{2}, \; xy = 1\right)$

39. A, C

Sol. Let M(h, k) be the image SM cuts a tangent at a point which lies on the auxiliary circle of the ellipse



$P \equiv \left(\frac{h \pm 4}{2}; \frac{k+0}{2}\right)$

Equation of auxiliary circle is $x^2 + y^2 = 25$

$\left(\frac{h \pm 4}{2}\right)^2 + \frac{k^2}{4} = 25$

40. B, C

Sol. Equation of tangent to the parabola $y^2 = 32x$

$y = mx + \frac{8}{m}$ ..... (1)

Which is also touches the hyperbola $9x^2 - 9y^2 = 8$

i.e., $x^2 - y^2 = \frac{8}{9}$

So, $\left(\frac{8}{m}\right)^2 = \left(\frac{8}{9}\right)m^2 - \frac{8}{9}$ by solving $m = \pm 3$ from equation (1), we get

$y = 3x + \frac{8}{3} \Rightarrow y = -3x - \frac{8}{3}$

41. C

Sol. $\lim\limits_{t \to x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$, using L'Hospital's Rule $\frac{-f'(x)\sin x + f(x)\cos x}{\sin^2 x} = 1$

$\Rightarrow \quad f(x) = \left(\frac{\pi}{3} - x\right)\sin x \quad \because \quad f\left(\frac{\pi}{6}\right) = \frac{\pi}{12}$, also $\sin x > x - \frac{x^3}{6}$ gives $\left(\frac{\pi}{3} - x\right)\sin x < x - \frac{x^3}{6}$ for $x \in \left(\frac{\pi}{3}, \pi\right)$

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42. C, D

Sol. $S_1 = \frac{1}{n}\sum_{r=1}^{n-1} \frac{\frac{r}{n}}{1+\left(\frac{r}{n}\right)^2}$; $S_2 = \frac{1}{n}\sum_{r=2}^{n} \frac{\frac{r}{n}}{1+\left(\frac{r}{n}\right)^2}$

Let $f(x) = \frac{x}{1+x^2}$; $f'(x) = \frac{1-x^2}{\left(1+x^2\right)^2}$

$\therefore$ f(x) is increasing in (0, 1) $\Rightarrow S_1 < \int_0^1 f(x)dx$; $S_2 > \int_0^1 f(x)dx$

43. C

Sol. Curves are $y^2 = 4a(x-a)$, $(x-4a)^2 + y^2 = 9a^2$

Equation of tangent for parabola $y = m(x-a) + \frac{a}{m}$

$\Rightarrow \left|\frac{4am + \frac{a}{m} - am}{\sqrt{m^2+1}}\right| = 3a \Rightarrow \left|3am + \frac{a}{m}\right| = 3a\sqrt{1+m^2} \Rightarrow m = \pm\frac{1}{\sqrt{3}}$

Equation of common tangents are $y = \pm\frac{1}{\sqrt{3}}(x-a) \pm a\sqrt{3}$

44. B

Sol. $A_1 = \begin{bmatrix}2 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + \begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = I + B \Rightarrow A^n = (I+B)^n = {}^nC_0 I + {}^nC_1 B + ... + {}^nC_n B^n$

Now, $B^2 = 2I \Rightarrow B^{2k} = 2^k I$ and $B^{2k+1} = 2^k B$

Now, $\begin{bmatrix}2 & 1\\ 1 & 0\end{bmatrix}^n = \left(C_0 + C_2 \cdot 2 + C_4 \cdot 2^2 + ...\right)I + \left(C_1 + C_3 \cdot 2 + ...\right)B$

$= XI + YB = \begin{bmatrix}X & O\\ O & X\end{bmatrix} + \begin{bmatrix}Y & Y\\ Y & -Y\end{bmatrix} = \begin{bmatrix}X+Y & Y\\ Y & X-Y\end{bmatrix}$

$\Rightarrow a_{12} = {}^nC_1 + {}^nC_3\, 2 + {}^nC_5 2^2 + ... = \frac{1}{\sqrt{2}}\left[\frac{\left(1+\sqrt{2}\right)^2 - \left(1-\sqrt{2}\right)^n}{2}\right]$

Similarly, $a_{22} = \frac{\left(1+\sqrt{2}\right)^n + \left(1-\sqrt{2}\right)^n}{2} - \frac{\left(1+\sqrt{2}\right)^n - \left(1-\sqrt{2}\right)^n}{2\sqrt{2}} = \frac{\left(\sqrt{2}-1\right)\left(1+\sqrt{2}\right)^n - \left(\sqrt{2}+1\right)\left(1-\sqrt{2}\right)^n}{2\sqrt{2}}$

Now, $\lim_{n\to\infty} \frac{a_{12}}{a_{22}} = 1+\sqrt{2} \Rightarrow \ell^2 = \sqrt{9} + \sqrt{8}$

45. D

Sol. (I) $\because$ (2, 3) lies on ax + by – 5 = 0

2a + 3b = 5 ..... (1)

$\because$ Lines is at greatest distances from centre $\left(\frac{4-3}{3-2}\right)\left(\frac{-a}{b}\right) = -1 \Rightarrow a = b$

(II) Let P be (h, k) then $h^2 + k^2 + 2h + 2k = 0$ ..... (2)

Midpoint of OP is $\left(\frac{h}{2}, \frac{k}{2}\right)$. Let $x = \frac{h}{2}, y = \frac{k}{2}$; h = 2x, k = 2y

$4x^2 + 4y^2 + 4x + 4y = 0$ from equation (1)

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(III) Centre of circles are $C_1 \equiv (1, 2)$, $C_2 \equiv (5, -6)$

Equation of C_1C_2 is $y - 2 = \frac{-6-2}{5-1}(x-1)$

$2x + y - 4 = 0$ ..... (1)

Equation of radical axis is $8x - 16y - 56 = 0$ ..... (2)

It of intersection of equation (1) and (2) is $(3, -2)$

(IV) Let $S \equiv x^2 + y^2 - 6\sqrt{3}x - 6y + 27 = 0$

Equation of pair of tangents is $SS_1 = T^2$

$18y\left(y - \sqrt{3}x\right) = 0$

46. C

Sol. (I) $f(x) = \int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\,dt + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\,dt$ = constant

$\Rightarrow f'(x) = 0 \Rightarrow f(x) = c \Rightarrow f\left(\frac{\pi}{4}\right) = \frac{\pi}{4} \Rightarrow f(x) = \frac{\pi}{4}$

(II) $\left.\frac{d(f(x))}{d(\ln x)}\right|_{x=\ln 2} = e^{6\ln 2} - e^{4\ln 2} = 2^6 - 2^4 = 48$

(III) $\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx = -\frac{\pi}{2}\ln 2$

(IV) $y = \frac{1}{\left(\frac{1}{x+1}\right)^2 + \frac{1}{x+1} - 2} \Rightarrow x \neq -1,\ \frac{1}{x+1} \neq -2, 1$

Section – B

47. 8.00

Sol. $5\left(x^3 - 5x^2 + 6x\right) - 20\left({}^{x-1}C_5\right) - 4\left({}^{x-1}C_4\right) + 16\left({}^{x-1}C_3\right) > 0$

$\Rightarrow 20\left({}^{x-1}C_5 + {}^{x-1}C_4\right) - 16\left({}^{x-1}C_4 + {}^{x-1}C_3\right) < 5\left(x^3 - 5x^2 + 6x\right)$

$\Rightarrow 20\left({}^{x}C_5\right) - 16\left({}^{x}C_4\right) < 5\left(x^3 - 5x^2 + 6x\right)$

$\Rightarrow 4\left({}^{x-1}C_4\right) - 4\left({}^{x-1}C_3\right) < 5(x-2)(x-3)$

$\Rightarrow 4(x-1)(x-2)(x-3)\left(\frac{x-4}{24} - \frac{1}{6}\right) < 5(x-2)(x-3)$

$\Rightarrow (x - 2)(x - 3)(x - 11)(x + 2) < 0$

$\Rightarrow x = 6, 7, 8, 9, 10$

48. 7.71

Sol. Number of ways $= {}^9C_4 + 5\left({}^9C_3\right) + 6\left({}^9C_2\right) + {}^9C_1 = 771$

49. 15.00

Sol. $a = 3,\ b = 2,\ e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{4}{9} = \frac{5}{9}$

$e = \frac{\sqrt{5}}{3}$ $f_1 = \sqrt{5}, f_2 = -\sqrt{5}$

$P_1 : y^2 = 4\sqrt{5}x,\ P_2 : y^2 = -12\sqrt{5}x$

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$T_1 : y = m_1x + \frac{3}{m_1}$, $T_2 : y = m_2x - \frac{3\sqrt{5}}{m_2}$

T_1 passes through $(3f_2, 0)$ and T_2 passes through $(f_1, 0)$

$0 = -3\sqrt{5}m_1 + \frac{\sqrt{5}}{m_1}$ $\qquad$ $0 = \sqrt{5}m_2 - \frac{3\sqrt{5}}{m_2}$

$\Rightarrow \frac{\sqrt{5}}{m_1} = 3\sqrt{5}m_1$ $\qquad$ $\sqrt{5}m_2 = \frac{3\sqrt{5}}{m_2}$

$\Rightarrow m_1^2 = \frac{1}{\sqrt{5}}$ $\qquad$ $m_2^2 = 3$

$\frac{2}{m_1^2} + 3m_2^2 = 6 + 9 = 15$

50. 5.00

Sol. SA $\perp$ BC

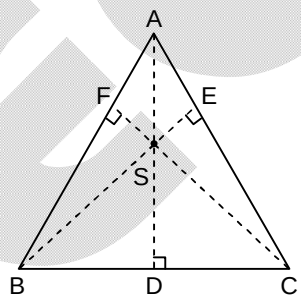
$\left(\frac{2at_1 - 0}{at_1^2 - a}\right)\left(\frac{2at_3 - 2at_2}{at_3^2 - at_2^2}\right) = -1$

$\Rightarrow 4t_1 + t_1^2(t_2 + t_3) = t_2 + t_3$ ..... (1)

Similarly $4t_2 + t_2^2(t_3 + t_1) = t_3 + t_1$ ..... (2)

On subtracting (2) from (1), we get

$|t_1t_2 + t_2t_3 + t_3t_1| = 5$



51. 1.00

Sol. $P_1 : 4x + 7y + 4z + 81 = 0$ ..... (1)

$P_2 : 5x + 3y + 10z = 25$ ..... (2)

Equation of plane passing through their line of intersection is

$(4x + 7y + 4z + 81) + \lambda(5x + 3y + 10z - 25) = 0$

$\Rightarrow (4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + 81 - 25\lambda = 0$ ..... (3)

Equation (3) $\perp$ (1)

$\therefore \lambda = -1$

Equation of plane is $-x + 4y - 6z + 106 = 0$

52. 3.14

Sol. Domain $\Rightarrow x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \{0, 1, -1\}$

53. 1.00

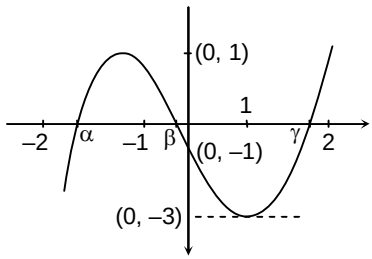
Sol. $f(x) = x^3 - 3x - 1$

$f(f(x)) = 0$

Let $f(x) = t$

$t = \alpha, \beta, \gamma$ for $x \in (2, \infty)$

$f(f(x)) = 0$ will have only one solution



54. 2023.00

Sol. f(x) is a periodic function with period = 2

$\therefore f(x + 2k) = f(x)$

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